Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

2 h 30 min · 100 marks · 11–12 long questions (7–14 marks) · calculator · 1.5 min per mark. Every paper: transformations with matrices + a money/percentage context. Most: Venn, histogram, vectors, simultaneous equations, graphs or kinematics.

FORMULAE
Often printed beside the question (sine/cosine rule, inverse matrix, quadratic formula, cone volume) — but learn them anyway.

TOP 12 MARK-LOSERS

- Answer only where algebra is required = **0**.
- Two transformations named in one description.
- Enlargement without "scale factor" or centre.
- MN applied in the wrong order (N is first).
- Image not labelled.
- Tangent "proof" without discriminant = 0 and conclusion.
- Vector ratio guessed, no coefficients equated.
- Angle without its reason.
- Graph solution with no line drawn.
- Wrong bounds for "definitely enough".
- Currencies compared without converting both.
- Probability quadratic: impossible n not rejected.

MONEY AND BOUNDS

- 378 in 5 : 3 : 1 ⇒ 42 per part.
- Profit share: 20% + 80% × 55/80 = 75%.
- Reverse: 5.10 ÷ 1.0625 = 4.80 · 15.95 ÷ 1.16 = 13.75.
- Compound: $20\,000 \times 1.084^2 = 23\,501.12$.
- Currency: same currency, include tax + postage, state cheaper.
- 48% of 2.35×10^7, then 5 : 3 : 2.

BOUNDS
Max area/volume: all upper bounds. Max a/b = a_{max}/b_{min}.
"Definitely enough paint": $area_{max}$ vs coverage.
Rod that definitely fits: diagonal with lower bounds.

ALGEBRA

SIMULTANEOUS
$x^2 + y^2 = 26$, $2x + y = 9 \Rightarrow 5x^2 - 36x + 55 = 0 \Rightarrow$ (5, −1), (2.2, 4.6).
Tangent: $x^2 + y^2 = 64$, $4y + 3x = 40 \Rightarrow 25x^2 - 240x + 576 = 0$, disc = 0 ⇒ one point (4.8, 6.4).

- $2^x4^y = 128 \Rightarrow x + 2y = 7$; $8^x/4^y = 32 \Rightarrow 3x - 2y = 5 \Rightarrow$ (3, 2).
- $x^{2/3} - 3x^{1/3} - 10 = 0 \Rightarrow y = x^{1/3} \Rightarrow x = 125, -8$.
- $25x^2 - 36 = (5x - 6)(5x + 6)$; ÷ = × reciprocal.
- $c = 5(f - 32)/9 \Rightarrow f = 9c/5 + 32$.
- $a \propto 1/b$, $b \propto 1/c^3 \Rightarrow a \propto c^3$.
- f = px + q from two points; ff(x) = p(px + q) + q.

INEQUALITIES AND GRAPHS

- $x < 3.5$ and $-6 \le x \le 1 \Rightarrow -6 \le x \le 1$.
- Region: solid for ≤/≥, dashed for ; label R; list integer points.
- Plot: table → points → smooth curve.
- Draw a line: curve = line. $x^2 + 2x + 4/x = 5x + 3 \Rightarrow$ draw $y = 5x + 3$.
- Gradient condition: point + dy/dx ⇒ two equations.

MATRICES

TOOLKIT
det = ad − bc · inverse (1/det)(d −b / −c a)
$M(x\ y)^T = (p\ q)^T \Rightarrow (x\ y)^T = M^{-1}(p\ q)^T$
$BA = C \Rightarrow B = CA^{-1}$ · area factor = |det|

TRANSFORMATIONS

Transformation	Matrix
reflect x-axis	(1 0 / 0 −1)
reflect y-axis	(−1 0 / 0 1)
reflect y = x	(0 1 / 1 0)
reflect y = −x	(0 −1 / −1 0)
rot 90° anticlockwise	(0 −1 / 1 0)
rot 90° clockwise	(0 1 / −1 0)
rot 180°	(−1 0 / 0 −1)
enlarge SF k (O)	(k 0 / 0 k)

DESCRIBE FULLY (ONE ONLY)
Rotation: angle, direction, centre · Enlargement: SF, centre · Reflection: line · Translation: vector.
N: $N \times D = B \Rightarrow N = B\,D^{-1}$. MN: N first. Negative SF: opposite side, inverted.

CALCULUS AND KINEMATICS

- v = dx/dt, a = dv/dt; rest v = 0; reject t < 0.
- $s = 4t^3 - 18t^2 + 5t + 10 \Rightarrow a = 24t - 36 > 0 \Rightarrow t > 1.5$.
- $x = kt^2 - t^3$, $v(2) = 24 \Rightarrow k = 9$; rest $t = 6 \Rightarrow 108$ m.
- $S = 4x^2 + 3a/x$ min: $3a = 8x^3 \Rightarrow S = 12x^2 = 400 \Rightarrow x = 5.77$.
- Condition on v ⇒ quadratic inequality in t ⇒ a at the ends.

GEOMETRY

- Tangent–secant: $TA^2 = TC \times TD \Rightarrow r = 2.25$.
- Circle reasons: cyclic 180° · centre = 2 × circumference · alternate segment · tangent ⊥ radius.
- Kite: AC = 10, area 48, BE = 4.8; similar by 2 angles + reasons.
- Exact chord: semicircle 90° + Pythagoras, surd form.
- 3D: vertical right-angled triangle; printed cosine rule for others.
- Quadrilateral: split into triangles.

MENSURATION

- Hemisphere : total = 8 : 25, total 1660 ⇒ r = 6.33, cone h = 26.9.
- Packing: floor(edge ÷ length) along each edge, multiply.
- Sector rings: difference of sectors.
- Regular polygon: exterior 360/n; area = n × ½ab sin C.

VECTORS

METHOD

Routes in a and b; scalar for a point on a line; same vector two ways; equate coefficients.

Collinear: AP = k PB + common point → ratio m : n. Parallel: scalar multiple.

OA = 6a, OB = 4b, OC : OA = 2 : 3, CD ∥ AB ⇒ DB = (4/3)b.

STATISTICS

- Mean Σmf ÷ Σf; median class at n/2.
- FD = f ÷ width; probability from fractions of bars.
- Scale: 30-min bar = 3 cm square ⇒ 1 cm = 10 min.
- 34 × 49 − 11 × 72 = 874 ⇒ 874 ÷ 23 = 38.0.
- Pie angle = f ÷ total × 360.

PROBABILITY AND SETS

- Unknown n: expression = given value → quadratic → reject impossible. 11 black, w white, P(WW) = 10/21.
- Given all same colour: P(all red) ÷ P(all same).
- Expected = p × n; sum different days.
- Venn with x: centre → pairs − x → singles → total; greatest x when all regions ≥ 0.
- P(C | P) = n(C ∩ P) ÷ n(P).